

Differential Geometry

Exercise 7

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7.1 Problem (DoCarmo, Riemannian Geometry page 56 question 2)

Let X and Y be vector fields on a manifold M equipped with a connection ∇ . Let $p \in M$ and $\gamma: I \rightarrow M$ an integral curve of X through p , so $\gamma(t_0) = p$ and $\dot{\gamma}(t) = X_{\gamma(t)}$. Denote the parallel transport map by $P_{t_0 t_1}^\gamma$, show that

$$\nabla_X Y|_p = \left. \frac{d}{dt} \right|_{t=t_0} (P_{t_0 t}^\gamma)^{-1} (Y|_{\gamma(t)}).$$

First we prove a few basic claims regarding parallel transport (which I'm not sure were proven in class, I apologize).

Firstly, $P_{t_0 t_1}^\gamma: T_{\gamma(t_0)}M \rightarrow T_{\gamma(t_1)}M$ is linear. Indeed, if $v \in T_{\gamma(t_0)}M$ is a vector and $X \in \mathfrak{X}(\gamma)$ is the vector field parallel along γ such that $X(t_0) = v$, then for $c \in \mathbb{R}$, cX is parallel along γ and $cX(t_0) = cv$. Indeed, since D_t is $\mathbb{R}$ -linear we have $D_t(cX) = cD_tX = 0$. Thus $P_{t_0 t_1}^\gamma(cv) = (cX)(t_1) = cX(t_1) = cP_{t_0 t_1}^\gamma(v)$.

And similarly it is additive. If $v, u \in T_{\gamma(t_0)}M$ are vectors with respective parallel vector fields $X, Y \in \mathfrak{X}(\gamma)$ starting at v, u respectively, then $X + Y \in \mathfrak{X}(\gamma)$ is parallel along γ since $D_t(X + Y) = D_tX + D_tY = 0$, and $(X + Y)(t_0) = v + u$. Thus

$$P_{t_0 t_1}^\gamma(v + u) = (X + Y)(t_1) = X(t_1) + Y(t_1) = P_{t_0 t_1}^\gamma(v) + P_{t_0 t_1}^\gamma(u).$$

Also $P_{t_0 t_1}^\gamma$ is invertible with inverse $P_{t_1 t_0}^\gamma$. Indeed if $u \in T_{\gamma(t_1)}M$ has vector field $X \in \mathfrak{X}(\gamma)$ parallel such that $X(t_1) = u$, then $P_{t_1 t_0}^\gamma(u) = X(t_0) = v$. Since X is parallel along γ and $X(t_0) = v$, we have $P_{t_0 t_1}^\gamma(v) = X(t_1) = u$. Thus

$$P_{t_0 t_1}^\gamma \circ P_{t_1 t_0}^\gamma = \text{id},$$

and by symmetry the swapped composition is the identity as well, so they are inverses.

Now if $v \in T_{\gamma(t_0)}M$, let $X_v \in \mathfrak{X}(\gamma)$ be the unique vector field parallel along γ such that $X_v(t_0) = v$. That is, $X_v(t) = P_{t_0 t}^\gamma(v)$ by definition of the parallel transport map. Now if $b_1, \dots, b_n \in T_{\gamma(t_0)}M$ form a basis, then $X_{b_1}, \dots, X_{b_n}$ form a local frame of $\mathfrak{X}(\gamma)$ of parallel vector fields. They are a local frame since $X_{b_i}|_t = P_{t_0 t}^\gamma(b_i)$, and since $P_{t_0 t}^\gamma$ is a linear isomorphism it maps bases to bases, so $X_{b_i}|_t$ is a basis for each t . In particular there exists a local frame of $\mathfrak{X}(\gamma)$ of parallel vector fields.

Now getting to the task at hand, notice that by definition of the pullback connection,

$$D_t(Y \circ \gamma)|_t = \nabla_{\dot{\gamma}(t)} Y|_{\gamma(t)} = \nabla_{X \circ \gamma(t)} Y|_{\gamma(t)} = \nabla_X Y|_{\gamma(t)}.$$

Thus since $p = \gamma(t_0)$,

$$\nabla_X Y|_p = D_t(Y \circ \gamma)|_{t_0}.$$

Let $E_1, \dots, E_n$ be a local frame of $\mathfrak{X}(\gamma)$ of parallel vector fields (which exists as shown above). Since $Y \circ \gamma \in \mathfrak{X}(\gamma)$, there are smooth $Y^1, \dots, Y^n: I \rightarrow \mathbb{R}$ such that $Y \circ \gamma = Y^i E_i$. Then since $D_t(E_i) = 0$ we have

$$D_t(Y \circ \gamma) = D_t(Y^i E_i) = \dot{Y}^i E_i + Y^i D_t(E_i) = \dot{Y}^i E_i.$$

And since E_i are parallel,

$$P_{t_0 t}^\gamma(Y \circ \gamma(t)) = P_{t_0 t}^\gamma(Y^i(t) E_i(t)) = Y^i(t) P_{t_0 t}^\gamma(E_i(t)) = Y^i(t) E_i(t_0)$$

Putting everything together we have

$$\left. \frac{d}{dt} \right|_{t=t_0} P_{t_0 t}^\gamma(Y \circ \gamma(t)) = \left. \frac{d}{dt} \right|_{t=t_0} (Y^i(t) E_i(t_0)) = \dot{Y}^i(t_0) E_i(t_0) = D_t(Y \circ \gamma)|_{t_0} = \nabla_X Y|_p.$$

Keeping in mind that $(P_{t_0 t}^\gamma)^{-1} = P_{t t_0}^\gamma$, we have the desired result.

7.2 Problem

Let (M, g) be a Riemannian manifold and $S \subseteq M$ a compact submanifold.

- (1) Define the **normal subbundle** ν_S by

$$\nu_S = \bigsqcup_{p \in S} T_p S^\perp.$$

And denote $\pi: \nu_S \rightarrow S$ to be the obvious projection operator. Show that ν_S is a submanifold of TM and π is smooth.

- (2) Define $\exp: \nu_S \rightarrow M$ to be the restriction of the exponential operator to ν_S . Show that there is an open subset $U \subseteq \nu_S$ with $S \cong \{(p, 0) \in \nu_S \mid p \in S\} \subseteq U$ such that $\exp(U)$ is open and $\exp$ restricts to a diffeomorphism $U \rightarrow \exp(U)$. Thus $T = \exp(U) \subseteq M$ is an open neighborhood of S , and we have a retract $r: T \rightarrow S$ given by $r = \pi \circ (\exp|_U)^{-1}$. T is called a **tubular neighborhood**.
- (3) Let $f: M \rightarrow N$ be a proper submersion. Let $p \in N$ and $S = f^{-1}p$, so that $S \subseteq M$ is a compact submanifold since f is a proper submersion. Let T, r be as in the previous point. Show that there exists an open neighborhood $p \in V \subseteq N$ and $f^{-1}V \subseteq T$. Define $g: f^{-1}V \rightarrow V \times S$ by $q \mapsto (f(q), r(q))$. Show that after possibly shrinking V , g is a diffeomorphism, and that f is a locally trivial fibration.

- (1) The restriction of the tangent bundle $TM|_S = \bigsqcup_{p \in S} T_p M$ is a vector bundle over S , and $TS \subseteq TM|_S$ is a subbundle. Let $p \in S$ and let U be a neighborhood of p in S over which there is a local frame $E_1, \dots, E_k$ of TS . By potentially shrinking U we can extend this to a local frame $E_1, \dots, E_k, E_{k+1}, \dots, E_n$ of $TM|_S$. If we then apply the Gram-Schmidt process to the E_i s, they remain smooth (since the inner product of two vector fields is a smooth function), and we obtain an orthonormal local frame $E'_1, \dots, E'_n$. The Gram-Schmidt process doesn't change the linear span of $E'_1, \dots, E'_i$ for all i , so we see that $E'_1, \dots, E'_k$ form an orthonormal local frame for TS and $E'_{k+1}, \dots, E'_n$ span ν_S (over U).

Thus in some neighborhood around every $p \in S$ there are local sections $E'_{k+1}, \dots, E'_n$ which span ν_S at the points where they are defined. And thus ν_S is a subbundle of $TM|_S$, and since $TM|_S$ is embedded in TM , we see that ν_S is embedded in TM as well.

And the projection map $\pi: TM \rightarrow M$ restricts along its domain and codomain to $\pi: \nu_S \rightarrow S$. Since these are both embedded submanifolds, π remains smooth.

- (2) Let us denote by $\exp^M$ the exponential map (from a subset of) $TM \rightarrow M$, so $\exp$ is the restriction of $\exp^M$ to ν_S . And let $S_0 = \{(p, 0) \in \nu_S \mid p \in S\} \subseteq \nu_S$. Consider the zero section $\zeta: S \rightarrow \nu_S$ given by $p \mapsto 0 \in T_p S^\perp$, whose image is S_0 . This is smooth as the restriction of the zero vector field $M \rightarrow TM$ along embedded submanifolds. We have that $\pi \circ \zeta = \text{id}$, so ζ is a topological embedding as $\pi|_{S_0}$ is its continuous inverse from its image. And $d\pi_{(p,0)} \circ d\zeta_p = \text{id}$ for all $p \in S$ so $d\zeta_p$ is injective, thus ζ is an immersion. As an immersive topological embedding, it is a smooth embedding, and so its image, S_0 , is an embedded submanifold of S_0 . Since S is compact, so is S_0 and thus it is closed in ν_S .

Let $p \in S$, we want to show that $d(\exp)_{p,0}: T_{(p,0)}\nu_S \rightarrow T_p M$ is surjective. We consider $\exp$'s action on two embedded submanifolds of ν_S : S_0 and $T_p S^\perp$ (the latter is embedded as it is the level set of p under the submersion π). Notice that on S_0 , $\exp$ maps $(p, 0)$ to p and thus is the composition of the diffeomorphism $S_0 \rightarrow S$ with the embedding $S \hookrightarrow M$. In particular, $d\exp_{(p,0)}$ maps $T_{(p,0)}S_0$ isomorphically to $T_p S \subseteq T_p M$. And on $T_p S^\perp$, $\exp$ agrees with $\exp_p^M$ which is a diffeomorphism around 0. Thus $d\exp_{(p,0)}$ is equal to $d(\exp_p^M)_0$ and thus maps $T_{(p,0)}T_p S^\perp$ (viewed as a subspace of $T_0 T_p M \cong T_p M$) isomorphically to $T_p S^\perp$ (since $d(\exp_p^M)_0$ is the identified with the identity). Since $T_p M = T_p S \oplus T_p S^\perp$, we see that $d\exp_{(p,0)}$ is indeed surjective.

Now, μ_S is a vector bundle over S of rank $\dim M - \dim S$ (since this is the dimension of $T_p S^\perp$). Thus its dimension is $\dim \mu_S = \dim M - \dim S + \dim S = \dim M$. And so each $d\exp_{(p,0)}$ is an isomorphism as its domain and codomain have the same dimensions and it is surjective. So by the inverse function theorem, $\exp$ is a diffeomorphism from a neighborhood of $(p,0) \in U_p \subseteq \nu_S$ to a neighborhood of $p \in V_p \subseteq M$.

By potentially shrinking them, we can assume each U_p is a precompact coordinate ball. And since S is compact and the U_p cover it, we take a finite subcover $U_{p_1}, \dots, U_{p_n}$. Let $U_0 = \bigcup_i U_{p_i}$, which remains precompact as the finite union of precompact sets. And $\exp$ is a local diffeomorphism $U_0 \rightarrow \bigcup_p V_p = V_0$ which is open.

Now we make the following claim: if $A \subseteq X$ is closed in a compact Hausdorff space X , and $f: X \rightarrow Y$ is locally injective and injective on A , then there is a neighborhood U of A such that f is injective on U . Suppose not, then for every neighborhood U of A there are $x_U \neq y_U$ such that $f(x_U) = f(y_U)$. This forms two nets $x_\bullet, y_\bullet$, and since X is compact they have convergent subnets, $x_{U_i} \rightarrow x$ and $y_{U_i} \rightarrow y$ (we can assume they are over the same directed set). This means that for any neighborhood $x \in V$, eventually x_{U_i} is in V and in particular $V \cap U_i$ is non-empty for some i . Since the U_i are cofinal, this means that every neighborhood of x intersects every neighborhood of A . Since X is compact Hausdorff and A is closed, A is compact, and in Hausdorff spaces disjoint compact sets have disjoint neighborhoods. Thus x must be in A , and similarly y is in A .

Since f is continuous, $f(x_{U_i}) \rightarrow f(x)$ and $f(y_{U_i}) \rightarrow f(y)$ and since $f(x_{U_i}) = f(y_{U_i})$, we see that $f(x) = f(y)$ (by uniqueness of limits in a Hausdorff space). By assumption f is injective on A so $x = y$. And by local injectivity there is a neighborhood U of $x = y$ on which f is injective. Eventually x_{U_i}, y_{U_i} are in this neighborhood and since $f(x_{U_i}) = f(y_{U_i})$ we must have $x_{U_i} = y_{U_i}$, a contradiction. (This argument can be done using second-countability or metrizable and sequences instead of nets.)

Going back to our problem, we invoke the above result with $X = \bar{U}_0$, $A = S_0$, and $\exp$. So there is a neighborhood U of S_0 on which $\exp$ is injective. Thus $\exp: U \rightarrow \exp(U)$ is a bijective local diffeomorphism, and thus a diffeomorphism. Since $\exp: U_0 \rightarrow V_0$ is a local diffeomorphism to an open subset $V_0 \subseteq M$, it is in particular an open map and thus $\exp(U)$ is open.

And $r = \pi \circ (\exp|_U)^{-1}$ is indeed a retract $T \rightarrow S$ since it is smooth as the composition of smooth maps, and for $p \in S$, $(\exp|_U)^{-1}(p) = (p, 0)$ and thus $r(p) = p$ as desired.

(3) Let us define

$$g: T \rightarrow N \times S, \quad q \mapsto (f(q), r(q)).$$

Then relative to a product chart of $N \times S$, g 's differential is given by

$$dg_q = (df_q, dr_q).$$

So $\ker dg_q = \ker df_q \cap \ker dr_q$. And we know that since S is f 's level set, $\ker df_q = T_q S$ for all $q \in S$. And we have $r|_S = \text{id}_S$, so dr_q acts on $T_q S$ as the identity, thus $\ker df_q \cap \ker dr_q = 0$. So dg_q is injective for all $q \in S$, and thus g has full rank at q .

Furthermore, $\dim S = \dim M - \dim N$ and thus

$$\dim(N \times S) = \dim N + \dim S = \dim M = \dim T,$$

so that dg_q is an isomorphism for $q \in S$. By the inverse function theorem there is an open neighborhood $q \in U_p \subseteq T$ such that g restricts to a diffeomorphism onto its image on U_p . By the same argument as above (when restricting $\exp$ to a diffeomorphism around S_0), g restricts to a diffeomorphism on an open neighborhood $S \subseteq W \subseteq T$.

Now we claim that there is a neighborhood $p \in V$ such that $f^{-1}V \subseteq W$. Suppose not, so focusing on a coordinate chart around p , let $B_{1/n}$ be a coordinate ball (in this coordinate chart) of radius $1/n$ around p . Then $f^{-1}B_{1/n}$ is not a subset of W . So let $x_n \in f^{-1}B_{1/n} - W$ for each n . Since x_n are all contained in the compact $\bar{B}_1$, there is a convergent subsequence $x_{m_n} \rightarrow x$. By continuity, $f(x_{m_n}) \rightarrow f(x)$ and since $f(x_{m_n}) \in B_{1/m_n}$ which is decreasing, we see that $f(x) \in \bigcap_n B_{1/m_n} = \{p\}$ so that $f(x) = p$. Thus $x \in S$, and since W is a neighborhood of S and $x_{m_n} \rightarrow x$, we must have that eventually x_{m_n} is in W , contradicting our choice of x_n to not be in W .

So letting $p \in V$ be a neighborhood such that $f^{-1}V \subseteq W$, we restrict g to $f^{-1}V$. Since g 's restriction to W is a diffeomorphism onto its image, g 's restriction to $f^{-1}V$ is a diffeomorphism as well. Furthermore since $g(f^{-1}V) = V \times S$, we have that $g: f^{-1}V \rightarrow V \times S$ is a diffeomorphism.

And since

$$\text{proj}_V \circ g(q) = \text{proj}_V(f(q), r(q)) = f(q),$$

we have that with $F_p = S = f^{-1}p$ and the diffeomorphism given by g for all $p \in N$, f is a locally trivial fibration as desired.

7.3 Problem (DoCarmo, Riemannian Geometry page 80 question 3b)

Let G be a Lie group with Lie algebra $\mathfrak{g}$ equipped with a bi-invariant metric $\langle \bullet, \bullet \rangle$. Show that G 's geodesics starting at the identity are precisely its one-parameter subgroups.

Let $\gamma: \mathbb{R} \rightarrow G$ be the one-parameter subgroup of a left-invariant vector field $Y \in \mathfrak{g}$, so $\gamma'(t) = Y_{\gamma(t)}$. Then its acceleration is

$$D_t \gamma'|_t = \nabla_{\partial/\partial t}^\gamma(Y \circ \gamma)|_t = \nabla_{\gamma'(t)}(Y)|_{\gamma(t)} = \nabla_Y Y|_{\gamma(t)}.$$

So it is sufficient to show that $\nabla_Y Y = 0$ for all left-invariant vector fields $Y \in \mathfrak{g}$.

Koszul's formula gives

$$2\langle X, \nabla_Y Y \rangle = 2Y\langle X, Y \rangle - X\langle Y, Y \rangle + 2\langle Y, [X, Y] \rangle - \langle X, [Y, Y] \rangle.$$

Since $\langle \bullet, \bullet \rangle$ is left-invariant, $\langle X, Y \rangle, \langle Y, Y \rangle$ are constant and thus $Y\langle X, Y \rangle = X\langle Y, Y \rangle = 0$. Since $[Y, Y] = 0$ as well, we have

$$\langle X, \nabla_Y Y \rangle = \langle Y, [X, Y] \rangle.$$

Bi-invariance of the metric implies that for all left-invariant vector fields,

$$\langle [X, Y], Z \rangle = -\langle X, [Z, Y] \rangle.$$

And in particular

$$\langle Y, [X, Y] \rangle = -\langle [Y, Y], X \rangle = 0.$$

So all in all we have that $\langle X, \nabla_Y Y \rangle = 0$ for all left-invariant X . The Levi-Civita connection is left-invariant, so $\nabla_Y Y \in \mathfrak{g}$ and thus it must be zero. So we have shown that its one-parameter subgroup is a geodesic.

If γ is a geodesic starting at e , then $\gamma'(0) \in T_e G$ induces a left-invariant vector field Y . As shown above, Y 's one-parameter subgroup is a geodesic starting at e with initial velocity $Y_e = \gamma'(0)$. By uniqueness of geodesics, γ must be equal to Y 's one-parameter subgroup, as desired.

7.4 Problem

Let $f: X \rightarrow Z$ and $g: Y \rightarrow Z$ be smooth maps. We say that f and g are **transverse** to one another, and write $f \pitchfork g$, if for all $x \in X$ and $y \in Y$ such that $f(x) = g(y)$ we have $df_x(T_x X) + dg_y(T_y Y) = T_{f(x)} Z$.

- (1) If $f \pitchfork g$ are transverse to one another, show that they define a categorical pullback $X \times_Z Y$ in the category of smooth manifolds.
- (2) Let $k = \mathbb{R}, \mathbb{C}$, and $\pi_V: V \rightarrow M$ be a smooth vector bundle over k . Show that π_V is a submersion.
- (3) Let $f: N \rightarrow M$ be a smooth map, show that $N \times_M V$ is a smooth manifold.
- (4) Show that there is a smooth map $V \times_M V \rightarrow V$ given by fiberwise addition.

- (5) Let $\underline{k} \rightarrow M$ denote the trivial bundle $k \times M \rightarrow M$. Show that there is a smooth map $\underline{k} \times_M V \rightarrow V$ given by fiberwise scalar multiplication.

- (1) Let $\Delta \subseteq Z \times Z$ denote the diagonal. This is an embedded submanifold as the image of the injective embedding $Z \rightarrow Z \times Z$, $z \mapsto (z, z)$. This is clearly a topological embedding, and smooth, and it is an immersion since its differential is represented by the matrix $(\text{id}, \text{id})^\top$ in local coordinates (obtained by choosing a coordinate chart in Z , then the product of that coordinate chart with itself is a coordinate chart of its image under the diagonal map, and this matrix is the representation of the differential); which is of full rank.

So Δ is the injective image of a smooth embedding, and is thus an embedded submanifold of $Z \times Z$.

Let $f \times g: X \times Y \rightarrow Z \times Z$ be given coordinate-wise. Notice that

$$(f \times g)^{-1}\Delta = \{(x, y) \in X \times Y \mid (fx, gy) \in \Delta\} = \{(x, y) \mid fx = gy\} = X \times_Z Y.$$

So to show that $X \times_Z Y$ is an embedded submanifold of $X \times Y$ (and in particular a smooth manifold), it is sufficient to show that $f \times g$ is transverse to Δ .

Indeed, for $fx = gy$ we have, under the identification $T_{x,y}X \times Y \cong T_xX \times T_yY$,

$$d(f \times g)_{x,y}(T_{x,y}X \times Y) \cong df_x(T_xX) \times dg_y(T_yY).$$

Let $z = fx = gy$, and $T_{z,z}\Delta$ is then the diagonal of $T_{z,z}Z \times Z$. This is because if we let $\iota_\Delta: Z \rightarrow Z \times Z$ be the diagonal map, then $T_{z,z}\Delta = d(\iota_\Delta)_z T_z Z$. As explained, ι_Δ 's differential is the matrix $(\text{id}, \text{id})^\top$, which maps $v \in T_z Z$ to (v, v) . Thus $d(\iota_\Delta)_z T_z Z$ is indeed the diagonal of $T_z Z \times T_z Z \cong T_{z,z}Z \times Z$.

In general if we have vector subspaces $U, W \subseteq V$ such that $U + W = V$, then $U \times W + \Delta = V \times V$, where $\Delta \subseteq V \times V$ is the diagonal. For $(x, y) \in V \times V$ we want $u \in U, w \in W, v \in V$ such that $(u + v, w + v) = (x, y)$. This requires $u - w = x - y$; and since $U + W = V$, such u, w exist. Then let $v = x - u$ so $u + v = x$ and $w + v = u + y - x + x - u = y$ as desired.

In our case, $V = T_z Z$, $U = df_x(T_x)$, and $W = dg_y(T_y)$, so indeed we have that $f \times g$ is transverse to Δ and $X \times_Z Y$ is an embedded submanifold of $X \times Y$.

Since it is an embedded submanifold, we can restrict the projection maps $X \times Y \rightarrow X, Y$ to it. Thus the square

$$\begin{array}{ccc} X \times_Z Y & \xrightarrow{p_Y} & Y \\ p_X \downarrow & & \downarrow g \\ X & \xrightarrow{f} & Z \end{array}$$

Indeed commutes since

$$g \circ p_Y(x, y) = g(y) = f(x) = f \circ p_X(x, y).$$

And this indeed is a pullback: if $U \xrightarrow{h} X, U \xrightarrow{k} Y$ commute with f, g , to form $\ell: U \rightarrow X \times_Z Y$ we must have

$$\ell(u) = (h(u), k(u)).$$

(This can be shown directly, also because $X \times_Z Y$ is the pullback in the category of topological spaces.) This is smooth as a map to $X \times Y$, and since $X \times_Z Y$ is an embedded submanifold, we can restrict it to its image, which is $X \times_Z Y$, and it remains smooth. Thus ℓ is unique (since $X \times_Z Y$ is the pullback in the category of topological spaces) and it satisfies the property.

Since the pullback is a universal property; it is unique up to unique diffeomorphism. In particular, if we had another smooth structure on $X \times_Z Y$ which made it the pullback along with p_X, p_Y , then the ℓ defined above would be a diffeomorphism. But in this case

$$\ell(x, y) = \ell(p_X(x, y), p_Y(x, y)) = (x, y) \implies \ell = \text{id}.$$

So id would be a diffeomorphism, meaning they have the same smooth structure.

- (2) Let $\Phi: \pi_V^{-1}U \rightarrow U \times \mathbb{R}^n$ be a local trivialization, where $U \subseteq M$ is open. So it is a diffeomorphism and $\text{proj}_1 \circ \Phi = \pi_V$, where $\text{proj}_1: U \times \mathbb{R}^n \rightarrow U$ is the projection map onto the first factor. Thus for $v \in \pi_V^{-1}U$,

$$d(\pi_V)_v = d(\text{proj}_1)_{\Phi v} \circ d\Phi_v.$$

Since Φ is a diffeomorphism and proj_1 a submersion, $d(\pi_V)_v$ is surjective for all $v \in \pi_V^{-1}U$.

The local trivializations cover V (for $v \in V$, $\pi_V(v)$ is contained in an open subset U over which there is a local trivialization, and $v \in \pi_V^{-1}U$), so $d(\pi_V)_v$ is surjective for all $v \in V$. That means π_V is a submersion.

- (3) Submersions are transverse to every map: $d(\pi_V)_v(T_v V) = T_{\pi_V v} M$ since it is a submersion. In particular $d(\pi_V)_v(T_v V) + df_q(T_q N) = T_{\pi_V v} M$ whenever $f q = \pi_V v$. So $\pi_V \pitchfork f$, and we showed in the first point that this means $N \times_M V$ is a smooth manifold.
- (4) Suppose V has rank n . That is, its local trivializations are maps of the form $\Phi: \pi_V^{-1}U \rightarrow U \times \mathbb{R}^n$ and V has dimension $m + n$ where $m = \dim M$.

We know that $V \times_M V$'s codimension in $V \times V$ is equal to Δ 's (the diagonal's) codimension in $M \times M$ (by the first point). Since $\dim \Delta = \dim M = m$ this tells us

$$\dim(V \times V) - \dim(V \times_M V) = \dim(M \times M) - \dim \Delta \implies \dim(V \times_M V) = 2(m + n) - m = 2n + m.$$

And explicitly,

$$V \times_M V = \{(v, u) \in V \times V \mid \pi_V(v) = \pi_V(u)\} = \bigsqcup_{p \in M} V_p \times V_p,$$

where $V_p = \pi_V^{-1}p$. We want to show that

$$\text{add}: V \times_M V \rightarrow V, \quad (v, u) \mapsto v + u$$

is smooth. It is well-defined since V_p is a vector space by assumption, so we can add vectors in it.

Now, for $(v, u) \in V_p \times V_p$ they both lie in the domain of the same trivialization, say $\Phi: \bar{U} \rightarrow U \times \mathbb{R}^n$ where $\bar{U} = \pi_V^{-1}U$. Let us denote by $\Phi_\ell: \bar{U} \rightarrow \mathbb{R}^n$ Φ 's projection onto the second factor. Since $V \times_M V$ is embedded, $(\bar{U} \times \bar{U}) \cap (V \times_M V)$ is open in $V \times_M V$ (as it is the intersection with an open set). And define

$$\Psi: (\bar{U} \times \bar{U}) \cap (V \times_M V) \rightarrow U \times k^n \times k^n, \quad (v, u) \mapsto (\pi_V(v), \Phi_\ell(v), \Phi_\ell(u)).$$

This is a smooth map, as it is clearly smooth coming from the open set $(\bar{U} \times \bar{U}) \cap (V \times V)$ (as all the coordinates are smooth), and restricting to $V \times_M V$ is restricting to an embedded submanifold, and thus remains smooth.

Ψ is also bijective. It is injective since if $\Psi(v, u) = \Psi(v', u')$ then $\pi_V(v) = \pi_V(u) = \pi_V(v') = \pi_V(u') = p$. Then $\Phi_\ell(v) = \Phi_\ell(v')$, but $\Phi_\ell|_{V_p}$ is a linear isomorphism and thus injective, so $v = v'$. Similar for $u = u'$. And it is surjective: for $(p, a, b) \in U \times k^n \times k^n$, since $\Phi_\ell|_{V_p}$ is a linear isomorphism and thus surjective, there are $v, u \in V_p$ such that $\Phi_\ell(v) = a, \Phi_\ell(u) = b$. Then

$$\Psi(v, u) = (\pi_V(v), \Phi_\ell(v), \Phi_\ell(u)) = (p, a, b)$$

as desired.

Finally, notice that (choosing a product chart)

$$d\Psi = (d\pi_V, d\Phi_\ell, d\Phi_\ell).$$

Since Φ is a diffeomorphism and Φ_ℓ is its projection onto the first factor, it is the composition of submersions (Φ with projection), and thus a submersion. So π_V, Φ_ℓ are all submersions, and thus of ranks m, n respectively. Thus $d\Psi$ has rank $2m + n = \dim U \times k^{2n}$, so Ψ is an immersion and submersion, thus a local diffeomorphism. It is also bijective, and thus a diffeomorphism.

Now consider the composition

$$U \times k^n \times k^n \xrightarrow{\Psi^{-1}} V \times_M V \xrightarrow{\text{add}} V \xrightarrow{\Phi} U \times k^n, \quad (p, a, b) \mapsto (p, a + b).$$

This is clearly smooth (addition in k^n is smooth). Since Φ, Ψ are diffeomorphisms, this means that **add** is smooth when restricted to $\overline{U} \times \overline{U}$. Since local trivializations cover V , such open sets $\overline{U} \times \overline{U}$ cover $V \times_M V$, and thus **add** is smooth as desired.

(5) Explicitly we have

$$\underline{k} \times_M V = \{(c, p, v) \in \underline{k} \times V \mid \pi_V(v) = p\}.$$

And we want to show that

$$\text{mult}: \underline{k} \times_M V \rightarrow V, \quad (c, p, v) \mapsto cv$$

is smooth.

Notice that $\underline{k} \times_M V$ is diffeomorphic to $k \times V$. Define

$$\begin{aligned} F: \underline{k} \times_M V &\rightarrow k \times V, & (c, p, v) &\mapsto (c, v), \\ G: k \times V &\rightarrow \underline{k} \times_M V, & (c, v) &\mapsto (c, \pi_V(v), v). \end{aligned}$$

These are both smooth, since F is the restriction of the obvious map $\underline{k} \times V \rightarrow k \times V$ (given by the same formula) which is smooth (as its coordinates are) to an embedded submanifold on its domain. And G is the restriction of the obvious map $k \times V \rightarrow \underline{k} \times V$ (given by the same formula), which is smooth as its coordinates are, to an embedded submanifold on its codomain.

And F, G are inverses:

$$F \circ G(c, v) = F(c, \pi_V(v), v) = (c, v), \quad G \circ F(c, p, v) = G(c, v) = (c, \pi_V(v), v) = (c, p, v).$$

Thus they form diffeomorphisms.

So to show **mult** is smooth it is sufficient to show $\text{mult} \circ G: k \times V \rightarrow V$ is smooth. We identify **mult** with $G \circ \text{mult}$, so it is given by

$$\text{mult}(c, v) = cv.$$

Let $\Phi: \overline{U} \rightarrow U \times k^n$ be a local trivialization. Then consider the composition

$$k \times U \times k^n \xrightarrow{\text{id} \times \Phi^{-1}} k \times \overline{U} \xrightarrow{\text{mult}} \overline{U} \xrightarrow{\Phi} U \times k^n.$$

(The target of **mult** remains $\overline{U}$ since for $v \in \overline{U} = \pi_V^{-1}U$ we have $\pi_V(cv) = \pi_V(v) = p \in U$ so $cv \in \overline{U}$.) This is given by

$$(c, p, a) \mapsto (c, \Phi^{-1}(p, a)) \mapsto c\Phi^{-1}(p, a) \mapsto (p, ca).$$

The final mapping is because Φ is a linear isomorphism on the fibers, so

$$\Phi(c\Phi^{-1}(p, a)) = c(p, a) = (p, ca).$$

Multiplication over k is smooth, so this map is smooth.

Thus **mult** is smooth over $k \times \overline{U}$, and since local trivializations cover V , it is smooth over all of $k \times V$. And as explained, this means the original **mult**, defined on $\underline{k} \times_M V$, is smooth.